

Liver disease

The safety and efficacy of Nevirapine has not been established in patients with significant underlying liver disorders. Nevirapine is contraindicated in patients with severe hepatic impairment (see section 4.3). Patients with chronic hepatitis B or C and treated with combination antiretroviral therapy are at an increased risk for severe and potentially fatal hepatic adverse events. In the case of concomitant antiviral therapy for hepatitis B or C, please refer also to the relevant product information for these medicinal products.

Patients with pre-existing liver dysfunction including chronic active hepatitis have an increased frequency of liver function abnormalities during combination antiretroviral therapy and should be monitored according to standard practice. If there is evidence of worsening liver disease in such patients, interruption or discontinuation of treatment must be considered.

Post-exposure-prophylaxis

Serious hepatotoxicity, including liver failure requiring transplantation, has been reported in HIV-uninfected individuals receiving multiple doses of Nevirapine in the setting of post-exposure-prophylaxis (PEP), an unapproved use. The use of Nevirapine has not been evaluated within a specific study on PEP, especially in term of treatment duration and therefore, is strongly discouraged.

Other warnings

Combination therapy with Nevirapine is not a curative treatment of patients infected with HIV-1; patients may continue to experience illnesses associated with advanced HIV-1 infection, including opportunistic infections.

The long-term effects of nevirapine are unknown at this time. Combination therapy with Nevirapine has not been shown to reduce the risk of transmission of HIV-1 to others through sexual contact or contaminated blood.

Combination antiretroviral therapy has been associated with the redistribution of body fat (lipodystrophy) in HIV infected patients. The long-term consequences of these events are currently unknown. Knowledge about the mechanism is incomplete. A connection between visceral lipomatosis and PIs and lipotrophy and NRTIs has been hypothesised. A higher risk of lipodystrophy has been associated with individual factors such as older age, and with drug related factors such as longer duration of antiretroviral treatment and associated metabolic disturbances. Clinical examination should include evaluation for physical signs of fat redistribution. Consideration should be given to the measurement of fasting serum lipids and blood glucose. Lipid disorders should be managed as clinically appropriate.

Nevirapine may interact with some medicinal products; therefore, patients should be advised to report to their doctor the use of any other medications.

Oral contraceptives and other hormonal methods of birth control should not be used as the sole method of contraception in women taking Nevirapine, since nevirapine might lower the plasma concentrations of these medications. For this reason, and to reduce the risk of HIV transmission, barrier contraception (e.g. condoms) is recommended. Additionally, when oral contraceptives are used for hormonal regulation during administration of Nevirapine the therapeutic effect should be monitored.

Pharmacokinetic results suggest caution should be exercised when Nevirapine is administered to patients with moderate hepatic dysfunction and should not be administered in patients with severe hepatic dysfunction. Overall, the results suggest that patients with mild to moderate hepatic dysfunction, defined as Child-Pugh Classification Score ≤ 7 , do not require an adjustment in Nevirapine dosing. In patients with renal dysfunction, who are undergoing dialysis, pharmacokinetic results suggest that supplementing Nevirapine therapy with an additional 200 mg dose of Nevirapine following each dialysis treatment would help offset the effects of dialysis on nevirapine clearance. Otherwise patients with CLcr 20 ml/min do not require an adjustment in Nevirapine dosing.

Immune reactivation syndrome

In HIV-infected patients with severe immune deficiency at the time of institution of combination antiretroviral therapy (CART), an inflammatory reaction to asymptomatic or residual opportunistic pathogens may arise and cause serious clinical conditions, or aggravation of symptoms. Typically, such reactions have been observed within the first few weeks or months of initiation of CART. Relevant examples are cytomegalovirus retinitis, generalised and/or focal mycobacterial infections, and Pneumocystis carinii pneumonia. Any inflammatory symptoms should be evaluated and treatment instituted when necessary.

Drug interactions:

The interaction data are presented as percent changes (geometric mean) with a 95% prediction interval (95% PI).

NRTIs: No dosage adjustments are required when Nevirapine is taken in combination with zidovudine, didanosine, or zalcitabine. When zidovudine data were analyzed from a study (n = 11) in which HIV-1 infected patients received Nevirapine 400 mg/day in combination with 100-200 mg t.i.d. of zidovudine, nevirapine resulted in a mean decrease of 24% in zidovudine AUC (95% PI: -69%, +83%) and a mean decrease of 26% in zidovudine Cmax (95% PI: -84%, +254%). The clinical relevance of the decrease in zidovudine plasma levels is currently unknown. Paired data suggest that zidovudine had no effect on the pharmacokinetics of nevirapine. In one crossover study, nevirapine had no clinically relevant effect on the steady-state pharmacokinetics of either didanosine (n = 18) or zalcitabine (n = 6).

Results from a 28 day study in HIV infected patients (n = 22) administered Nevirapine, neftinavir (750 mg t.i.d.) and stavudine (30-40 mg b.i.d.) showed no clinically relevant changes in the AUC or Cmax of stavudine. Furthermore, a population pharmacokinetic study of 90 patients assigned to receive lamivudine with Nevirapine or placebo revealed no changes to lamivudine apparent clearance and volume of distribution, suggesting no induction effect of nevirapine on lamivudine clearance.

Non-nucleoside reverse transcriptase inhibitors (NNRTIs): Limited pharmacokinetic data (n=17) are available on the co-administration of nevirapine and efavirenz. No major pharmacokinetic interactions were observed, only a limited decrease in efavirenz exposure was observed. However, this co-administration is not recommended since the co- administration of efavirenz and nevirapine could lead to a higher risk for side effects. Moreover this co-administration does not improve efficacy over either NNRTI alone.

PIs: Nevirapine is a mild to moderate inducer of the hepatic enzyme CYP3A; therefore, it is possible that co-administration with PIs (also metabolised by CYP3A) may result in an alteration in the plasma concentration of either agent.

Saquinavir: The limited data available with saquinavir soft gel capsule boosted with ritonavir do not suggest any clinically relevant interaction between saquinavir boosted with ritonavir and nevirapine.

Results from a clinical trial (n = 19) with HIV infected patients administered Nevirapine and indinavir (800 mg q8h) indicated that their co-administration leads to a mean decrease of 31% in indinavir AUC (95% PI: -64%, +30%), a mean decrease of 15% in Cmax (95% PI: -53%, +55%), and a mean decrease of 44% in Cmin (95% PI: -77%, +39%). No clinically relevant change in nevirapine plasma levels was found. No definitive clinical conclusions have been reached regarding the potential impact of co-administration of nevirapine and indinavir. A dose increase of indinavir to 1000 mg q8h should be considered when indinavir is given with nevirapine 200 mg b.i.d.; however, there are no data currently available to establish that the short term or long term antiviral activity of indinavir 1000 mg q8h with nevirapine 200 mg b.i.d. will differ from that of indinavir 800 mg q8h with nevirapine 200 mg b.i.d.

Results from a clinical trial (n = 18) with HIV infected patients administered Nevirapine and ritonavir (600 mg b.i.d.) indicated that their coadministration leads to no clinically relevant change in ritonavir or nevirapine plasma levels.

Results from a 28 day study in HIV infected patients (n = 23) administered Nevirapine, stavudine (30-40 mg b.i.d.) and neftinavir (750 mg t.i.d.) showed no clinically relevant changes in neftinavir pharmacokinetic parameters after the addition of nevirapine. Compared to historical controls nevirapine levels appeared to be decreased. The major metabolite of neftinavir (AG1402) decreased when administered with nevirapine. The overall effect of nevirapine on the total exposure of neftinavir plus the AG1402 metabolite was a mean decrease in AUC by 20% (95% PI: -72%, +128%), a mean decrease of 12% in Cmax (95% PI: -61%, +100%), and a mean decrease of 35% in Cmin (95% PI: -90%, +316%).

There were no increased safety concerns noted with the coadministration of Nevirapine with any of these PIs when used in combination.

In HIV positive adults, nevirapine used in combination with lopinavir/ritonavir 400/100 mg (3 capsules) twice daily has been reported to result in a decline in the lopinavir AUC of 27% compared with historical data. Although the clinical relevance of this observation has not been fully established, an increase in the dose of lopinavir/ritonavir to 533/133 mg (4 capsules) twice daily with food is recommended in combination with nevirapine.

Results from a pharmacokinetic study in paediatric patients were consistent with the findings in adults. During nevirapine co-administration, lopinavir AUC decreased by 22% (AUC ratio 0.78; 0.56-1.09) and lopinavir Cmin by 55% (Cmin ratio 0.45; 0.25-0.82). For children, increase of the dose of lopinavir/ritonavir to 300/75 mg/m² should be considered when used in combination with nevirapine, particularly for patients in whom reduced susceptibility to lopinavir/ritonavir is suspected.

Ketoconazole: Co-administration of ketoconazole (400 mg q.d.) following full induction of nevirapine resulted in a reduction of ketoconazole exposure at steady state. Ketoconazole AUC decreased by a mean of 72% (95% PI: -95%, +101%) and Cmax decreased by a mean of 44% (95% PI: -86%, +158%). In the same study (n = 21), ketoconazole administration resulted in a 15-28% increase in the plasma levels of nevirapine compared to historical controls. Ketoconazole and Nevirapine should not be given concomitantly. A similar decrease in itraconazole concentrations cannot be excluded.

Fluconazole: Co-administration of fluconazole and Nevirapine resulted in approximately 100% increase in nevirapine exposure compared with historical data where Nevirapine was administered alone. Because of the risk of increased exposure to nevirapine, caution should be exercised if the medicinal products are given concomitantly and patients should be monitored closely. There was no clinically relevant effect of nevirapine on fluconazole.

Oral Contraceptives: Nevirapine 200 mg b.i.d. was co-administered with a single dose of an oral contraceptive containing ethinyl estradiol (EE) 0.035 mg and norethindrone (NET) 1.0 mg. Nevirapine following 28 days of exposure (full induction) decreased the mean EE AUC by 20% (95% PI: -57%, +52%). Nevirapine also decreased the mean NET AUC by 19% (95% PI: -50%, +30%) and the Cmax by 16% (95% PI: -49%, +37%). Appropriate doses for oral contraceptives in combination with nevirapine, with respect to safety and efficacy, have not been established.

Other medicinal products metabolised by CYP3A: Nevirapine is an inducer of CYP3A and potentially CYP2B6, with maximal induction occurring within 2-4 weeks of initiating multiple-dose therapy. In presence of nevirapine, a 3-fold increase in clearance of oral methadone was observed. Methadone mean AUC and mean Cmax, corrected for the change in methadone dose administered, are decreased by 65% (95% PI: -82%, -32%) and 50% (95% PI: -67%, -25%), respectively. Narcotic withdrawal syndrome has been reported in patients treated with Nevirapine and methadone concomitantly. Methadone-maintained patients beginning Nevirapine therapy should be monitored for evidence of withdrawal and methadone dose should be adjusted accordingly.

Other compounds that are substrates of CYP3A and CYP2B6 may have decreased plasma concentrations when co-administered with Nevirapine. Therefore, careful monitoring of the therapeutic effectiveness of P450 metabolised medicinal products is recommended when taken in combination with Nevirapine.

CYP isoenzyme inhibitors: The results of a nevirapine-clarithromycin interaction study (n = 15) revealed a reduction in clarithromycin mean AUC by 31% (95% PI: -57%, +9%) and mean Cmin by 56% (95% PI: -92%, +126%); the active clarithromycin metabolite 14-OH clarithromycin showed an increase in the mean AUC by 42% (95% PI: -41%, +242%) and mean Cmax by 47% (95% PI: -39%, +255%). There were increases in the nevirapine Cmin by 28%, AUC by 26% and Cmax by 24% compared to historical controls. These results would suggest that no dose adjustment is necessary for either clarithromycin or Nevirapine when the two medicinal products are co-administered. Close monitoring of hepatic abnormalities is nevertheless recommended. However, alternative therapy to clarithromycin should be considered when treating a patient for mycobacterium avium-intracellulare complex, as the active metabolite is not effective in this instance.

Monitoring of steady-state nevirapine trough plasma concentrations in patients who received long-term Nevirapine treatment revealed that nevirapine trough concentrations were elevated in patients who received cimetidine (+ 7%, n = 13).

CYP isoenzyme inducers: An open-label study (n = 14) to determine the effects of nevirapine on the steady state pharmacokinetics of rifampicin resulted in no significant change in rifampicin Cmax and AUC. In contrast, rifampicin produced a significant lowering of nevirapine AUC (-58%), Cmax (- 50%) and Cmin (- 68%) compared to historical data.

The available pharmacokinetic data suggest that the concomitant use of rifampicin and Nevirapine is not recommended. Therefore, these medicinal products should not be used in combination. Physicians needing to treat patients co-infected with tuberculosis and using a Nevirapine containing regimen may consider use of rifabutin instead. Rifabutin and Nevirapine can be administered concurrently without dose adjustments (see below). Alternatively physicians may consider switching to a triple NRTI combination for a variable period of time, depending on the tuberculosis treatment regimen.

In a pharmacokinetic study (n = 19) the concomitant administration of rifabutin following full induction with Nevirapine resulted in a mean increase of 17% in the steady state AUC (95% PI: - 53%, +191%) and a 28% increase in mean Cmax (95% PI: -44%, +195%) of rifabutin. There was also an increase in the 25-O-desacetyl-rifabutin metabolite mean AUC by 24% (95% PI: -83%, +787%) and an increase in mean Cmax by 29% (95% PI: -67%, +400%). Due to the high intersubject variability some patients may experience large increases in rifabutin exposure and may be at higher risk for rifabutin toxicity. Therefore, caution should be used in concomitant administration. A clinically not relevant increase in the apparent clearance of nevirapine (by 9%) compared to historical pharmacokinetic data was reported.

Warfarin: The interaction between nevirapine and the antithrombotic agent warfarin is complex, with the potential for both increases and decreases in coagulation time when used concomitantly. The net effect of the interaction may change during the first weeks of co-administration or upon discontinuation of Nevirapine, and close monitoring of anticoagulation levels is therefore warranted.

Hypericum perforatum: Serum levels of nevirapine can be reduced by concomitant use of the herbal preparation St John's wort (Hypericum perforatum). This is due to induction of drug metabolism enzymes and/or transport proteins by St John's Wort. Herbal preparations containing St John's Wort should therefore not be combined with Nevirapine. If patient is already taking St John's Wort check nevirapine and if possible viral levels and stop St John's Wort. Nevirapine levels may increase on stopping St John's Wort. The dose of Nevirapine may need adjusting. The inducing effect may persist for at least 2 weeks after cessation of treatment with St John's Wort.

Other information: Studies using human liver microsomes indicated that the formation of nevirapine hydroxylated metabolites was not affected by the presence of dapsone, rifabutin, rifampicin, and trimethoprim/sulfamethoxazole. Ketoconazole and erythromycin significantly inhibited the formation of nevirapine hydroxylated metabolites.

Side effects: The most frequently reported adverse events related to Nevirapine therapy, across all clinical trials, were rash, nausea, fatigue, fever, headache, vomiting, diarrhoea, abdominal pain and myalgia.

The postmarketing experience has shown that the most serious adverse reactions are Stevens-Johnson syndrome and toxic epidermal necrolysis and serious hepatitis/hepatic failure and hypersensitivity reactions, characterised by rash with constitutional symptoms such as fever, arthralgia, myalgia and lymphadenopathy, plus visceral involvement, such as hepatitis, eosinophilia, granulocytopenia, and renal dysfunction. The first 18 weeks of treatment is a critical period which requires close monitoring.

The following adverse events which may be causally related to the administration of Nevirapine have been reported. The frequencies estimated are based on pooled clinical trial data for events considered related to Nevirapine treatment. Frequency classes: very common ($>1/10$); common ($>1/100$, $<1/10$); uncommon ($>1/1,000$, $<1/100$); rare ($>1/10,000$, $<1/1,000$); very rare ($<1/10,000$)

Blood and lymphatic system disorders

rare granulocytopenia, anaemia

Immune system disorders

common allergic reactions
rare hypersensitivity (syndrome), anaphylaxis

Nervous system disorders

common headache

Gastrointestinal disorders

common nausea
uncommon vomiting, abdominal pain
rare diarrhoea

Hepato-biliary disorders

Common hepatitis (1.2 %), liver function tests
uncommon abnormal jaundice
rare liver failure / fulminant hepatitis

Skin and subcutaneous tissue disorders

common rash (9 %)
uncommon Stevens Johnson syndrome (0.3 %),urticaria
rare toxic epidermal necrolysis, angio-oedema

Musculoskeletal, connective tissue and bone disorders

uncommon myalgia
rare arthralgia

General disorders and administration site conditions

uncommon fatigue, fever

Combination antiretroviral therapy has been associated with redistribution of body fat (lipodystrophy) in HIV infected patients including the loss of peripheral and facial subcutaneous fat, increased intra-abdominal and visceral fat, breast hypertrophy and dorsocervical fat accumulation (buffalo hump).

Combination antiretroviral therapy has been associated with metabolic abnormalities such as hypertriglyceridaemia, hypercholesterolaemia, insulin resistance, hyperglycaemia and hyperlactataemia.

The following events have also been reported when Nevirapine has been used in combination with other anti-retroviral agents: pancreatitis, peripheral neuropathy and thrombocytopenia. These events are commonly associated with other antiretroviral agents and may be expected to occur when Nevirapine is used in combination with other agents; however it is unlikely that these events are due to Nevirapine treatment. Hepatic-renal failure syndromes have been rarely reported.

In HIV-infected patients with severe immune deficiency at the time of initiation of combination antiretroviral therapy (CART), an inflammatory reaction to asymptomatic or residual opportunistic infections may arise.

Skin and subcutaneous tissues

The most common clinical toxicity of Nevirapine is rash, with v attributable rash occurring in 9 % of patients in combination regimens in controlled studies. In these clinical trials 24 % of patients treated with a Nevirapine containing regimen experienced rash compared with 15 % of patients treated in control groups. Severe rash occurred in 1.7 % of Nevirapine -treated patients compared with 0.2 % of patients treated in the control groups.

Rashes are usually mild to moderate, maculopapular erythematous cutaneous eruptions, with or without pruritus, located on the trunk, face and extremities. Allergic reactions (anaphylaxis, angioedema and urticaria) have been reported. Rashes occur alone or in the context of hypersensitivity reactions, characterised by rash associated with constitutional symptoms such as fever, arthralgia, myalgia and lymphadenopathy plus visceral involvement such as hepatitis, eosinophilia, granulocytopenia and renal dysfunction.

Severe and life-threatening skin reactions have occurred in patients treated with Nevirapine, including Stevens-Johnson syndrome (SJS) and, toxic epidermal necrolysis (TEN). Fatal cases of SJS, TEN and hypersensitivity reactions have been reported. The majority of severe rashes occurred within the first 6 weeks of treatment and some required hospitalisation, with one patient requiring surgical intervention.

Hepato-biliary

The most frequently observed laboratory test abnormalities are elevations in liver function tests (LFTs), including ALAT, ASAT, GGT, total bilirubin and alkaline phosphatase. Asymptomatic elevations of GGT levels are the most frequent. Cases of jaundice have been reported. Cases of hepatitis (severe and life-threatening hepatotoxicity, including fatal fulminant hepatitis) have been reported in patients treated with Nevirapine. In a large clinical trial, the risk of a serious hepatic event among 1121 patients receiving Nevirapine for a median duration of greater than one year was 1.2 % (versus 0.6 % in placebo group). The best predictor of a serious hepatic event was elevated baseline liver function tests. The first 18 weeks of treatment is a critical period which requires close monitoring.

Paediatric patients

Based on experience of 361 paediatric patients treated in clinical trials, the most frequently reported adverse events related to Nevirapine were similar to those observed in adults, with the exception of granulocytopenia which was more commonly observed in children. In post-marketing surveillance anaemia has been more commonly observed in children. Isolated cases of Stevens-Johnson syndrome or Stevens-Johnson/toxic epidermal necrolysis transition syndrome have been reported in this population.

Pregnancy and Lactation:

No observable teratogenicity was detected in reproductive studies performed in pregnant rats and rabbits. There are no adequate and well-controlled studies in pregnant women. Therefore Nevirapine should only be used during pregnancy if the expected benefit justifies the possible risk to the child and caution should be exercised when prescribing Nevirapine to pregnant women. Results from a pharmacokinetic study (ACTG 250) of 10 HIV-1 infected pregnant women who were administered a single oral dose of 100 or 200 mg Nevirapine at a median of 5.8 hours before delivery, have shown that nevirapine readily crosses the placenta and is found in breast milk. It is recommended that HIV infected mothers do not breast-feed their infants to avoid risking postnatal transmission of HIV and that mothers should discontinue nursing if they are receiving Nevirapine.

Signs and Symptoms of Over dose and treatment:

There is no known antidote for Nevirapine overdose. Cases of Nevirapine overdose at doses ranging from 800 to 6000 mg per day for up to 15 days have been reported. Patients have experienced oedema, erythema nodosum, fatigue, fever, headache, insomnia, nausea, pulmonary infiltrates, rash, vertigo, vomiting, increase in transaminases and weight decrease. All of these effects subsided following discontinuation of Nevirapine.

Storage Condition

Do not store above 30°C. Store in the original package.

Shelf Life:

36 months. The product should be used within 2 months of opening.

Presentation:

White opaque high-density polyethylene bottle, containing 100 ml and 240 ml of oral solution, with a plastic cap and polypropylene child resistant closure with wad having induction sealing liner. A 20 ml measuring cup is included in the pack.

Manufactured by:

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